Two constants in the numerical estimates of inter-universal Teichmüller theory IV

$12/(\ell(\ell+1))$ and $12/\ell^2$: a fifth report

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Abstract

The constants $\chi = 12/(\ell(\ell+1))$ and $\tau = 12/\ell^2$ occur at different steps of the numerical estimates surrounding IUT IV. The first is the exact ratio of the stage-one coefficient $1/(2\ell)$ to the mean of $j^2/(2\ell)$ over $j = 1, \dots, (\ell-1)/2$, and appears in Scholze–Stix's approximate calculation. The second is used in the upper-bound rearrangement in the proof of IUT IV, Theorem 1.10. We locate both in the original texts and compute their difference with exact rational arithmetic. With all other inputs fixed, the latter rearrangement adds $H/(2\ell^2)$ to the unnormalized expression, or $1/\ell$ after division by $H/(2\ell)$; its reciprocal factor is strictly larger. Tables at $\ell = 7, 157$ also give the contributions from the designated split primes of two explicit curves. These curves serve only as numerical illustrations; none of the algebraic propositions depends on their initial-data admissibility. What is compared here are two scalar expressions under a common set of inputs; this is not a comparison of the two sources' full estimates, whose positive terms, divisor supports and approximation conventions differ and are kept separate throughout. These calculations do not determine a possible-image hull or adjudicate IUT or the Scholze–Stix objection.

1 The objects and the comparison being made

The coefficient identity in [Fourth, §7] is the starting point. Here we identify a larger sufficient constant used in an upper-bound estimate, compare it with the sharp threshold for the scalar rearrangement defined below, and quantify the difference both before and after the source's root normalization. The finite-sum identity is elementary and is not claimed as a new discovery. Nor is priority claimed for comparing the two constants: the purpose is to record the exact source steps, common-input conditions and numerical differences in one place.

We reserve [S] for the explicitly quoted text or mathematical display of a source; [P] marks the deductions proved here, including the comparison of the displayed expressions. A source assertion and a proof of that assertion are different things. The hypotheses of the algebraic comparison below are written out; their satisfaction by an actual possible-image hull is not supplied by the arithmetic checks. All references to pages of IUT I and IV are to the consulted author versions, May 2020 and April 2020 respectively. Publication metadata are listed separately. For Scholze–Stix we use the manuscript of July 16, 2018.

Write H for a normalized degree of an *unrooted* Tate-parameter divisor and set

$$m = \frac{\ell-1}{2}, \qquad r = \frac{1}{2\ell}, \qquad Q = rH, \qquad b = \frac{\ell+1}{24}, \qquad c = \frac{\ell+1}{4}. \tag{1}$$

Until a source application is specified, these are real or rational quantities, with ℓ an odd integer at least 5 and $H > 0$. In IUT IV the relevant H is $\log(\mathfrak{q})$, the normalized degree of the divisor supported on the *selected* bad places [IUT4, Thm. 1.10, pp. 22–23]. The prose normalization there gives the following equality in our notation:

$$\text{[P]} \qquad |\log(\underline{\underline{q}})| = \frac{1}{2\ell}\log(\mathfrak{q}) \qquad \text{[IUT4, Thm. 1.10, p. 23].} \tag{2}$$

Here the double-underlined $\underline{\underline{q}}$ is the source's pilot notation; it is not an additional Tate parameter to which a second root should be applied.

Scholze–Stix define q_E by summing over *all* finite places of multiplicative reduction, and use $\deg(q_E)/(2\ell)$ for the rooted quantity [SS, §1.2, p. 2; §1.3, pp. 3–4]. Consequently, using the same elliptic curve is not alone enough to identify their $\deg(q_E)$ with IUT IV's selected H. Equality requires the same support, or explicit accounting for the omitted contribution. Under the extra good-reduction hypothesis of Theorem 1.10, the selected and actual bad places agree outside 2ℓ; the contribution over 2ℓ must still be checked. Likewise the positive terms appearing in the two texts are not identified merely by giving them the same symbol.

2 What is printed in Scholze–Stix

Their setup in §1.2 fixes a degree bound for k, restricts the local data at 2 and infinity, and uses split semistable reduction. The prime $\ell = \ell(P)$ is also constrained by the height window (1.3), the mod-ℓ image, and the local Tate extension condition; an exceptional list is set aside [SS, §1.2, pp. 2–3]. The substitutions in §5 evaluate coefficients, and do not assert these additional conditions for either displayed level.

[S] Their (1.4), p. 3 [SS, p. 3, (1.4); p. 4, (1.5)], is

$$-|\log(\underline{\underline{q}})| \leq -|\log(\underline{\underline{\Theta}})|. \tag{3}$$

On p. 4 the introduction to (1.5) reads "The right side of (1.4) is in essential approximation". The immediately following parenthesis defines $\mathfrak{d}(P)_v$ as the v-part of the logarithmic root discriminant. Retaining the source's ℓ^*, the displayed average is

$$\text{[S]} \qquad \begin{aligned} -|\log(\underline{\underline{\Theta}})| &\approx \frac{1}{\ell^*}\sum_{v}\sum_{j=1}^{\ell^*}\left(j\cdot\mathfrak{d}(P)_v - j^2\frac{1}{2\ell[k:\mathbb{Q}]}v(q_v)\log N(v)\right) \\ &= \frac{\ell^*+1}{2}\left(\mathfrak{d}(P) - \frac{1}{6}\deg(q_E)\right). \end{aligned} \tag{4}$$

The intermediate sum-of-squares evaluation in their display is omitted; no approximation sign is replaced by an equality. In our deductions $m = \ell^* = (\ell-1)/2$. The following *unnumbered* display is

$$\text{[S]} \qquad \frac{1}{6}\deg(q_E) \lessapprox \left(1 - \frac{12}{\ell(\ell+1)}\right)^{-1}\cdot\mathfrak{d}(P). \tag{5}$$

The next sentence says the error is handled "with a linear term in ℓ on the right side". It then states that this achieves Claim 6; the following sentence begins the authors' objection to the reasoning for (1.5), proposing (1.6) with the j^2 factor missing. We are quoting the approximate calculation they describe, not attributing their endorsement to it. Thus (5) must not be quoted as an error-free inequality with $\leq$, nor called their equation (1.5). The exact bound proved below is instead conditional on an explicitly specified additive error.

Proposition 1 (the two finite moments; [P]). *For odd $\ell \geq 5$,*

$$\frac{1}{m}\sum_{j=1}^{m} j = c, \qquad \frac{1}{m}\sum_{j=1}^{m} \frac{j^2}{2\ell} = b = \frac{c}{6}, \qquad \frac{b}{r} = \frac{\ell(\ell+1)}{12}.$$

In particular the unique number t for which $bt = r$ is

$$\chi = \frac{r}{b} = \frac{12}{\ell(\ell+1)}, \qquad A = \frac{1}{\chi} = \frac{\ell(\ell+1)}{12}. \tag{6}$$

Proof. The sums are $m(m+1)/2$ and $m(m+1)(2m+1)/6$. Substituting $m = (\ell-1)/2$ gives $c = (\ell+1)/4$ and $b = (m+1)(2m+1)/(12\ell) = (\ell+1)/24$. Dividing r by this positive b gives (6). $\square$

This explains both $(\ell+1)/4$ and $1/6$ in the algebra of (4): c is the first moment and $1/6 = b/c$. SS also discuss the geometric origin of $1/6$ in their height comparison (1.2), p. 2 [SS, p. 2, (1.2)]: the respective degrees there are $1/12$ and $1/2$. That discussion is subject to their bounded local data; it does not assert an exact equality between every height and $\deg(q_E)/6$. Their displayed sums and rearrangement account for the denominator $\ell(\ell+1)$. Our comparison with the ℓ^2 denominator uses the specific IUT IV estimate below, without attributing an unstated motive or claiming a first occurrence of this comparison.

3 The sufficient constant used in IUT IV

[P] Reading the negative terms in Theorem 1.10, proof, Step (v), p. 29 [IUT4, Thm. 1.10, proof, Step (v), p. 29], gives the following equality of negative terms in the first two lines of its procession-normalized upper bound (with $\ell^* = (\ell-1)/2$ as in the source):

$$-\frac{(2\ell^*+1)(\ell^*+1)}{12\ell}\log(\mathfrak{q}_{v_{\mathbb{Q}}}) = -\frac{\ell+1}{24}\log(\mathfrak{q}_{v_{\mathbb{Q}}}).$$

The same display has different term $\frac{\ell+5}{4}\log(\mathfrak{d}^K_{v_{\mathbb{Q}}})$. This is the average of $j+1$, whereas (4) averages j. [S] *The same rearrangement, with the same slack, is in print elsewhere.* [JoshiIV, p. 70] writes: "Hence one may replace the term $-\frac{1}{6}\cdot\log(q)$ in (6.11.3) by $-\frac{1}{6}\cdot\left(1-\frac{12}{\ell^2}\right)\cdot\log(q)$ and still have an excess contribution greater than $\frac{1}{2\ell}\log(q)$." No priority is claimed here for the rearrangement or for its slack; what this report adds is the exact comparison with χ of §2. *This report takes no position on the further claims of* [JoshiIV].

The next upper bound factors out $(\ell+1)/4$ and replaces the remaining positive coefficients by larger ones. For example,

$$\text{[P]} \qquad \frac{\ell+5}{4} = \frac{\ell+1}{4}\left(1+\frac{4}{\ell+1}\right) \leq \frac{\ell+1}{4}\left(1+\frac{4}{\ell}\right).$$

Thus the common negative coefficient does not make the full positive parts of the two sources equal.

The occurrence of $12/\ell^2$ can be located independently of this interpretation. [S] The statement of Theorem 1.10, p. 23 [IUT4, Thm. 1.10, p. 23; proof, Step (viii), p. 30], contains $-\frac{1}{6}(1-12/\ell^2)\log(\mathfrak{q})$ in its displayed choice of C_Θ. The proof, Step (viii), p. 30, ends the explanation of that upper-bound display with the complete estimate clause:

> [S] "— where we apply the estimate $\frac{\ell+1}{4}\cdot\frac{1}{6}\cdot\frac{12}{\ell^2} \geq \frac{1}{2\ell}$ [cf. the fact that $\ell \geq 1$]."

Let
$$\tau = \frac{12}{\ell^2}, \qquad u = b\tau = \frac{\ell+1}{2\ell^2}, \qquad \epsilon = u - r = \frac{1}{2\ell^2}.$$
The following identity pinpoints the change in the upper bound.

Proposition 2 (the rearrangement and its slack; [P])**.** *For $H > 0$ and any real D, define the numerical expression*
$$K_t(H,D) = c\left(D - \frac{1-t}{6}H\right) - rH.$$
Then
$$\begin{aligned} K_\chi(H,D) &= cD - bH, \\ K_\tau(H,D) &= cD - bH + \frac{H}{2\ell^2}, \\ \frac{K_\tau(H,D) - K_\chi(H,D)}{Q} &= \frac{1}{\ell}. \end{aligned}$$
Among $t \in \mathbb{R}$, the inequality $K_t(H,D) \geq cD - bH$ holds if and only if $t \geq \chi$.

Proof. Subtracting $cD - bH$ leaves $(bt - r)H$. At $t = \chi$ it is zero. At $t = \tau$ it is $((\ell+1)/(2\ell^2) - 1/(2\ell))H = H/(2\ell^2)$. Division by $Q = H/(2\ell)$ gives $1/\ell$. Since $b, H > 0$, its sign is the sign of $t - \chi$. □

In Step (viii) the terms represented by D include different, conductor and other positive estimates; a further prime-counting estimate follows on pp. 30–31. Proposition 2 compares the expressions with *these other terms held fixed.* In particular, $1/\ell$ is the change of this normalized numerical expression, not a measured change in the actual C_Θ or in a possible-image hull.

4 Comparison of the two reciprocal scalar factors

The comparison below is between two *scalar expressions* evaluated on a common set of inputs, which Proposition 3 states as a hypothesis. It is not a comparison of the full estimates of [SS] and [IUT4]: no map identifying their supports, positive terms, error terms or approximation conventions is given here, and none is claimed.

Proposition 3 (an explicit error and common inputs; [P])**.** *Let $H > 0$, let D, R be real, and put $U = D + R/c \geq 0$. Suppose a real number h satisfies the two bounds*
$$-rH \leq h \leq c(D - H/6) + R.$$
Then
$$\frac{H}{6} \leq F_\chi U \leq F_\tau U, \qquad F_\chi = \frac{1}{1-\chi}, \quad F_\tau = \frac{1}{1-\tau}. \tag{7}$$
The second inequality is strict when $U > 0$. The exact differences are
$$\tau - \chi = \frac{12}{\ell^2(\ell+1)}, \qquad \frac{\tau}{\chi} - 1 = \frac{1}{\ell}, \tag{8}$$
$$F_\tau - F_\chi = \frac{12\ell}{(\ell^2-12)(\ell(\ell+1)-12)}, \qquad \frac{F_\tau}{F_\chi} - 1 = \frac{12}{(\ell+1)(\ell^2-12)}. \tag{9}$$

Proof. The two assumed bounds give $(b-r)H \le cD + R$. As $b = c/6$ and $r = b\chi$, division by c gives $(1-\chi)H/6 \le U$. For odd $\ell \ge 5$, $0 < \chi < \tau < 1$, so $0 < F_\chi < F_\tau$. The displayed differences follow by subtraction and division of fractions. □

The hypothesis involving h is an explicit conditional comparison; the approximate symbol in SS is not silently replaced by it. For nonnegative common U, the χ bound is tighter. This orders the two *scalar forms*, not the complete inequalities of the two sources, whose positive terms and errors have not been equated. Although τ and χ have the same leading order $12/\ell^2$, their difference has order ℓ^{-3}. The same is true of the absolute and relative differences of the reciprocal factors in (9). The normalized expression difference in Proposition 2 instead has order ℓ^{-1}. These are different comparisons, with different denominators.

The permitted levels

The algebra above requires positive reciprocal denominators. $1-\chi > 0$ is equivalent to $\ell(\ell+1) > 12$, and $1-\tau > 0$ to $\ell^2 > 12$. At the odd prime $\ell = 3$ the first denominator vanishes and the second is negative. This is a boundary of the algebra, not a source counterexample.

[S] IUT I, Definition 3.1(c), p. 62 [IUT1, Def. 3.1(c), p. 62], begins: "ℓ is a prime number ≥ 5 such that the image of the outer homomorphism $G_F \to GL_2(\mathbb{F}_\ell)$ determined by the ℓ-torsion points of E_F contains the subgroup $SL_2(\mathbb{F}_\ell) \subseteq GL_2(\mathbb{F}_\ell)$;" The same item also requires ℓ to be prime to the selected residue characteristics and Tate orders. IUT IV, Theorem 1.10, p. 22, additionally assumes full 15-torsion over F and states $\ell \ne 5$. [P] The lower bound and primality therefore give $\ell \ge 7$. Indeed the proof's terminal estimates on p. 31 use that range and $d_{\mathrm{mod}} = [F_{\mathrm{mod}} : \mathbb{Q}] \ge 1$ [IUT4, Thm. 1.10, proof, Step (viii), p. 31]:

$$\text{[S]} \qquad (1 - 12/\ell^2)^{-1} \le 2, \qquad (1 - 12/\ell^2)^{-1}(1 + 12d_{\mathrm{mod}}/\ell) \le 1 + 20d_{\mathrm{mod}}/\ell.$$

These are the source's two estimates with its stated restrictions $\ell \ge 7$ and $d_{\mathrm{mod}} \ge 1$. At $\ell = 5, d_{\mathrm{mod}} = 1$ the second estimate fails, since its left side is $85/13 > 5$. Level 5 can illustrate the elementary formulas but not this theorem's terminal estimate. The range $\ell \ge 7$ by itself does not establish the remaining initial-data, reduction or lattice hypotheses of that theorem, including its precise torsion-generated choice of F on p. 22.

5 Exact values at two levels

All entries below are reduced rational numbers. The script compares the printed rows with independent finite-sum evaluations and the identities proved above. It does not convert an arithmetic parameter into admissible initial data.

ℓ	χ	τ	A	r	u	ϵ
7	$\frac{3}{14}$	$\frac{12}{49}$	$\frac{14}{3}$	$\frac{1}{14}$	$\frac{4}{49}$	$\frac{1}{98}$
157	$\frac{6}{12403}$	$\frac{12}{24649}$	$\frac{12403}{6}$	$\frac{1}{314}$	$\frac{79}{24649}$	$\frac{1}{49298}$

ℓ	F_χ	F_τ	$F_\tau - F_\chi$	$F_\tau/F_\chi - 1$
7	$\frac{14}{11}$	$\frac{49}{37}$	$\frac{21}{407}$	$\frac{3}{74}$
157	$\frac{12403}{12397}$	$\frac{24649}{24637}$	$\frac{942}{305424889}$	$\frac{6}{1946323}$

To connect these coefficients with explicit curves, use the first report's initial datum [First, Thm. 1 and the arithmetic conditions]:

$$L = \mathbb{Q}(\sqrt{5}), \quad a = (16 + 3\sqrt{5})^{29}, \quad E : y^2 = x(x-1)(x-a), \quad \ell = 7, \quad p = 211.$$

Its fields are $F = L(i, E[15])$ and $K = F(E[7])$. For comparison, the second report's section entitled *A curve carrying fourteen candidate levels* [Second, §13] uses

$$L = \mathbb{Q}(i), \quad a = (41 + 26i)^{472}, \quad E : y^2 = x(x-1)(x-a), \quad p = 2357,$$

and includes $\ell = 157$ among the candidates, with $F = L(E[15])$, $K = F(E[157])$. The latter section explicitly distinguishes candidate levels from a verification of all initial-data conditions at each level. We retain that distinction: the second row below is a coefficient evaluation at that explicit curve and level, not a new admissibility theorem.

In both designated quadratic fibres, one place is multiplicative and one good. At 211, the multiplicative reduction becomes split upon the unramified extension adjoining i, which leaves the Tate order unchanged; at 2357, it is already split over the indicated quadratic field. If $N = \mathrm{ord}_\pi(a)$, the unrooted Tate order at the bad place is $2N$, with $\mathrm{ord}_p(p) = 1$. Each base-field place has weight $1/2$, so

$$H_p = N \log p, \qquad Q_p = Nr \log p, \qquad (K_\tau - K_\chi)_p = N\epsilon \log p. \tag{10}$$

These are contributions from that rational prime only. No equality of the total selected degree with H_p is asserted. In particular, an extension's ramification index is not a second factor to insert into (10): the normalized degree is invariant under base extension [IUT4, Def. 1.9(i), pp. 21–22].

ℓ	p	N	$Q_p/\log p$	$bH_p/\log p$	$uH_p/\log p$	$\epsilon H_p/\log p$
7	211	29	$\frac{29}{14}$	$\frac{29}{3}$	$\frac{116}{49}$	$\frac{29}{98}$
157	2357	472	$\frac{236}{157}$	$\frac{9322}{3}$	$\frac{37288}{24649}$	$\frac{236}{24649}$

For example, the absolute difference of the reciprocal factors at $\ell = 157$ is $942/305424889$, while the designated-prime contribution to the unnormalized expression difference is $(236/24649)\log 2357$. After division by that fibre's Q_p, the latter is exactly $1/157$. A small difference of reciprocal factors and this normalized difference are not contradictory measurements; they answer different questions. None is an actual volume of a union of possible images.

Reproducibility and limits

Run `python3 scripts/constant_comparison.py` from the report bundle. It uses the standard-library `Fraction` type, evaluates both finite moments, checks the algebra and signs, and verifies all six printed table rows against the TeX source. It also requires the supplied frozen output file and checks exact agreement with the newly computed output. A successful run ends with `RESULT: PASS` and `EXIT:0`; failure returns a nonzero exit status. Running with Python optimization enabled is rejected. Finite numerical checks supplement the proofs; they do not prove the identification of source objects or compute omitted source error terms.

The exact comparison orders two scalar expressions under common inputs. It does not remove the approximation in SS, identify all its positive terms with IUT IV's, or apply a global lower bound to a single prime. In comments on the August 2018 SS manuscript, Mochizuki distinguishes the linear relationships between copies of $\mathbb{R}$ associated with prime-strips via the Θ-link from relationships between the log-volumes of regions before and after indeterminacies, which he describes as dependent on the region's geometry. For the latter relationship he writes: [S] "In particular, this relationship is highly non-linear." [Cmt, (C14), p. 4]. His report makes the same distinction and illustrates it by a region and its symmetrization [Rpt, §12, (MlLV), (LVEx), pp. 24–26]. These are attributed claims about the geometric comparison, not consequences of the scalar propositions above. Those propositions do not identify that comparison with multiplication by either reciprocal factor.

No conclusion is drawn about Corollary 3.12, the correctness of IUT, the validity of the SS objection, or *abc*. Large language models assisted drafting, source comparison and computations; the checks reported here are internal checks.

References

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